

BOTANY

1. Competitive inhibition of an enzyme occurs when an inhibitor molecule:

- A) Binds to an allosteric site, altering the enzyme's shape.
- B) Binds permanently to the active site, destroying it.
- C) Structurally resembles the substrate and competes for the active site.
- D) Denatures the enzyme by breaking peptide bonds.

2. Which mineral element is a central component of the chlorophyll molecule, essential for light absorption?

- A) Iron
- B) Calcium
- C) Magnesium
- D) Zinc

3. While reviewing scientific literature, you note the species name for the potato is written as **Solanum tuberosum Linn. What does 'Linn' indicate?**

- A) The generic name
- B) The specific epithet
- C) The family name
- D) The author who first described the species

4. Bacterial transformation, famously demonstrated by Frederick Griffith's experiments with mice, involves:

- A) Transfer of DNA via a viral bacteriophage vector.
- B) Uptake of naked DNA fragments from the surrounding environment.
- C) Direct physical contact and transfer of a plasmid via a pilus.
- D) Binary fission resulting in identical daughter cells.

5. In Phaeophyceae (Brown Algae), the primary reserve food materials are stored in the form of:

- A) Starch and cellulose
- B) Floridean starch
- C) Laminarin and mannitol
- D) Paramylon and glycogen

6. A student finds a mushroom in the forest. Structurally, the visible umbrella-like body is essentially the:

- A) Primary mycelium
- B) Gametangium
- C) Basidiocarp(fruiting body)
- D) Protonema

7. A defining difference in the life cycle of a Fern (*Pteridophyte*) compared to a Moss (*Bryophyte*) is that in ferns:

- A) The gametophyte is the dominant, photosynthetic phase.
- B) The sporophyte is the dominant, independent phase.
- C) Seeds are produced to protect the embryo.
- D) Water is no longer required for fertilization.

8. Which of the following features is distinctly characteristic of Angiosperms but generally absent in Gymnosperms?

- A) Presence of ovules
- B) Formation of pollen tubes
- C) Double fertilization and formation of triploid endosperm
- D) Heterosporous

9. A plant yields an economically important medicinal extract containing 'reserpine', widely used to treat hypertension. This plant is:

- A) *Azadirachta indica*
- B) *Rauwolfia serpentina*
- C) *Zingiber officinale*
- D) *Ocimum sanctum*

10. The protein coat that protects the nucleic acid core of a virus is called a:

- A) Capsid
- B) Envelope
- C) Pellicle
- D) Viroid

11. The unique tetradynamous condition of stamens (4 long inner, 2 short outer) is a diagnostic floral character of which family?

- A) Fabaceae
- B) Solanaceae
- C) Liliaceae
- D) Brassicaceae

12. [Applied] Which two gases released by industrial emissions are the primary contributors to the formation of acid rain?

- A) Carbon dioxide and Methane
- B) Sulfur dioxide and Nitrogen oxides
- C) Chlorofluorocarbons and Ozone
- D) Carbon monoxide and Ammonia

13. An epiphyte, like an orchid, grows on the branch of a large forest tree to access better sunlight. It does not extract nutrients from the tree. This interaction is classified as:

- A) Parasitism
- B) Mutualism
- C) Commensalism
- D) Amensalism

14. The primary greenhouse gas responsible for driving contemporary global climate change is:

- A) Ozone (O₃) B) Methane (CH₄) C) Nitrous Oxide (N₂O) D) Carbon Dioxide (CO₂)

15. In a freshwater pond ecosystem, organisms that live attached to or burrowing in the bottom sediments are collectively called:

- A) Plankton B) Nekton C) Benthos D) Neuston

16. A researcher extracts ribosomes from a bacterial cell and from the cytoplasm of a plant cell. What sedimentation coefficients will these respective ribosomes exhibit?

- A) Both will be 70S. B) Both will be 80S. C) Bacterial: 70S; Plant: 80S. D) Bacterial: 80S; Plant: 70S.

17. The primary function of the nucleolus within the eukaryotic nucleus is the:

- A) Synthesis of messenger RNA (mRNA) B) Synthesis and assembly of ribosomal RNA (rRNA)
C) Replication of genomic DNA D) Storage of lipid reserves

18. The network of endoplasmic reticulum that lacks attached ribosomes is called Smooth ER (SER). It is primarily involved in:

- A) Protein synthesis B) Lipid and steroid synthesis C) Intracellular digestion D) Photosynthesis

19. During the cell cycle, cells that have ceased dividing and entered a quiescent, inactive stage are said to be in the:

- A) G₁ phase B) G₂ phase C) G₀ phase D) M phase

20. Ergastic substances in a plant cell refer to:

- A) Active, division-promoting hormones B) Non-living cell inclusions like starch grains and crystals
C) Living cytoplasmic organelles D) Enzymes involved in respiration

21. A person has blood group AB. This expresses the phenotype of both A and B alleles equally. This phenomenon is a classic example of:

- A) Incomplete dominance B) Co-dominance C) Polygenic inheritance D) Pleiotropy

22. DNA replication is 'semi-conservative.' This means that in a newly synthesized DNA molecule:

- A) Both strands are entirely new.
B) One strand is original (parental) and one strand is newly synthesized.
C) Both strands are a patchwork mix of old and new DNA.
D) The DNA breaks into multiple short fragments that reassemble.

23. Down's syndrome in humans is a genetic disorder caused by the presence of an extra copy of which chromosome?

- A) Chromosome 13 B) Chromosome 18 C) Chromosome 21 D) Sex chromosomes (X/Y)

24. In genetic mapping, the distance between two linked genes on a chromosome is directly correlated to the:

- A) Size of the genes B) Percentage of crossing over (recombination frequency) between them
C) Number of mutations in the genes D) Ratio of dominant to recessive alleles

25. What is the primary role of transfer RNA (tRNA) during protein synthesis?

- A) It carries the genetic blueprint from DNA to the ribosome.
B) It acts as the primary structural component of the ribosome.
C) It acts as an adaptor molecule, bringing specific amino acids to the mRNA-ribosome complex.
D) It catalyzes the formation of peptide bonds.

26. The main enzyme responsible for transcribing DNA into mRNA is:

- A) DNA polymerase B) RNA polymerase C) Helicase D) Ligase

27. You are examining a root cross-section under a microscope. You observe a distinct barrel-shaped layer of cells with waxy, waterproof bands called Casparian strips. This layer is the:

- A) Epidermis B) Cortex C) Endodermis D) Pericycle

28. A distinguishing feature of a monocot root internal structure compared to a dicot root is the presence of:

- A) A very small or completely absent pith B) A well-developed, large central pith
C) Secondary growth through cambium activity D) Fewer than four xylem bundles (diarch/tetrarch)

29. Which essential conducting element of the phloem tissue lacks a true nucleus at maturity?

- A) Companion cells B) Phloem parenchyma C) Sieve tube elements D) Phloem fibres

30. A dry raisin swells up when placed in a bowl of water overnight. The primary physiological process driving this initial swelling is:

- A) Osmosis B) Imbibition C) Active transport D) Plasmolysis

31. In C₄ plants (like maize and sugarcane), carbon dioxide fixation occurs twice. The initial CO₂ fixation into oxaloacetic acid occurs in the:

- A) Bundle sheath cells B) Mesophyll cells C) Epidermal cells D) Guard cells

32. Photorespiration is a wasteful process in C₃ plants that significantly decreases photosynthetic efficiency. Which three organelles are involved in this pathway?

- A) Chloroplast, Mitochondria, Ribosome B) Chloroplast, Peroxisome, Mitochondria
C) Nucleus, Endoplasmic Reticulum, Golgi body D) Peroxisome, Lysosome, Vacuole

33. The active absorption of minerals by plant roots from the soil differs from passive absorption because active absorption strictly requires:

- A) A concentration gradient favoring diffusion B) The expenditure of metabolic energy (ATP)
C) The presence of a semi-permeable membrane D) High transpiration rates

34. A farmer notices that their shipped fruits are ripening too quickly and spoiling in transit. Which gaseous phytohormone is responsible for this rapid ripening effect?

- A) Auxin B) Cytokinin C) Gibberellin D) Ethylene

35. Which stage of cellular respiration is common to both aerobic and anaerobic pathways?

- A) Krebs Cycle B) Oxidative Phosphorylation C) Glycolysis D) Oxidative Decarboxylation

36. In the context of asexual reproduction, taking a cutting from a rose plant to grow a new plant relies on the process of:

- A) Sporogenesis B) Vegetative propagation C) Gametogenesis D) Apomixis

37. Pollination facilitated specifically by the agency of wind is scientifically termed:

- A) Entomophily B) Anemophily C) Hydrophily D) Ornithophily

38. A plant breeder crosses two distinct inbred lines of maize. The resulting F₁ offspring exhibit vastly superior height, yield, and disease resistance compared to both parents. This phenomenon is known as:

- A) Inbreeding depression B) Heterosis (Hybrid vigor) C) Polyploidy D) Mutation

39. In the context of tissue culture, the primary purpose of 'micropropagation' using shoot apical meristems is to:

- A) Produce plants that fix atmospheric nitrogen. B) Generate thousands of genetically identical, virus-free plants.
C) Cross two different species to create a hybrid. D) Extract secondary metabolites like alkaloids.

40. Golden Rice is a prominent example of bio-engineering designed to combat malnutrition. It has been genetically modified to produce high levels of:

- A) Vitamin C B) Iron and Zinc C) Beta-carotene (Pro-vitamin A) D) High-quality protein

ZOOLOGY

41. The wings of a butterfly and the wings of a bird both serve the function of flight, yet their anatomical structures and embryonic origins are entirely different. This is an example of:

- A) Homologous organs demonstrating divergent evolution. B) Vestigial organs demonstrating genetic drift.

C) Analogous organs demonstrating convergent evolution. D) Atavistic organs demonstrating a return to ancestral traits.

42. In the study of biochemical evidence for evolution, which of the following is considered the most compelling molecular proof of a universal common ancestor for all living organisms?

- A) All organisms utilize the exact same physical structures for locomotion.
- B) All organisms rely on the same universal genetic code (DNA/RNA) and fundamental metabolic pathways (like glycolysis).
- C) All organisms possess a multi-chambered heart.
- D) All organisms require environmental oxygen to survive.

43. Which concept represents the core mechanism of evolution proposed by Charles Darwin in his theory of Natural Selection?

- A) The inheritance of traits acquired during an organism's lifetime due to use or disuse.
- B) The sudden, large-scale mutation of an individual leading to a new species in one generation.
- C) Differential survival and reproduction of individuals based on heritable variations best suited to their environment.
- D) The purposeful drive of organisms towards biological perfection.

44. Sponges (Phylum Porifera) feed and respire through a unique anatomical feature called the canal system. What is the specific driving force that moves water continuously through this system?

- A) Muscular contractions of the sponge's body wall.
- B) The coordinated beating of flagella on specialized collar cells called choanocytes.
- C) Osmotic pressure differences between the ocean and the internal sponge cavity.
- D) External ocean currents simply flowing through passive tubes.

45. The Phylum Annelida represents a significant evolutionary advancement due to the presence of true metamerism. What does this term describe?

- A) The ability to alter color for camouflage.
- B) The division of the body into true linear segments, both externally and internally, allowing for coordinated movement.
- C) The presence of a hard, chitinous exoskeleton.
- D) The transition from radial to bilateral symmetry during larval development.

46. When examining a typical marine snail (Phylum Mollusca, Class Gastropoda), what is the function of the specialized feeding organ known as the radula?

- A) It is a venomous dart used to paralyze prey.
- B) It is a filter-feeding net extended into the water column.
- C) It is a ribbon-like structure covered in tiny chitinous teeth used to scrape food (like algae) off rocks.
- D) It is a muscular tube used to suck blood from host organisms.

47. Which defining characteristic is exclusively found in the Class Mammalia and is absent in all other vertebrate classes?

- A) The presence of a dorsal hollow nerve cord.
- B) The ability to regulate body temperature internally (endothermy).
- C) The production of milk from modified sweat glands to nourish their young.
- D) The development of lungs for terrestrial respiration.

48. Compact bone tissue is highly organized into structural units called Haversian systems (osteons). What is the critical function of the central Haversian canal?

- A) It stores yellow bone marrow for fat storage.
- B) It houses the blood vessels and nerves that supply the living bone cells (osteocytes).
- C) It provides a hollow space to make the bone lighter for movement.
- D) It acts as a joint capsule for synovial fluid.

49. The outermost layer of human skin (epidermis) is composed of stratified squamous epithelium. What structural modification makes this tissue highly effective at preventing water loss and resisting abrasion?

- A) The cells are heavily ciliated to sweep away debris.
- B) The outermost layers become heavily packed with the tough, waterproof protein keratin.

- C) The tissue contains massive amounts of highly elastic cartilage.
- D) The tissue is constantly bathed in highly alkaline mucus.

50. Areolar connective tissue is often described as the body's 'packing material'. Which of the following best describes its structure and function?

- A) It is composed tightly of parallel collagen fibers to provide unidirectional strength (like tendons).
- B) It is a loose matrix of fluid, varied fibers, and diverse cells that wraps organs, holding them in place while allowing for immune surveillance.
- C) It consists of cells filled completely with large lipid droplets for thermal insulation.
- D) It is a liquid tissue responsible for transporting dissolved gases.

51. In adipose tissue, the adipocytes are specialized to store triglycerides. Under a microscope, what distinguishing feature defines a mature white adipocyte?

- A) It possesses multiple nuclei arranged peripherally.
- B) The cytoplasm contains hundreds of tiny fat droplets surrounding a central nucleus.
- C) A single, massive fat droplet occupies almost the entire cell, pushing the nucleus flat against the plasma membrane.
- D) It contains large visible granules of glycogen.

52. During the erythrocytic schizogony of *Plasmodium* in human blood, what developmental stage immediately follows the 'ring stage'?

- A) Sporozoite B) Ookinete C) Amoeboid Trophozoite D) Gametocyte

53. Earthworms are monoecious (hermaphrodites), possessing both male and female reproductive organs. Why, then, do they practice cross-fertilization rather than self-fertilization?

- A) Their male and female organs are physically incompatible.
- B) They exhibit protandry, meaning the testes mature and produce sperm much earlier than the ovaries produce eggs.
- C) Self-fertilization requires water, which is absent inside the burrow.
- D) They lack the behavioral instinct to self-fertilize.

54. In the frog, the cloaca serves as a common exit chamber for which three physiological systems?

- A) Digestive, Respiratory, and Circulatory systems. B) Digestive, Excretory, and Reproductive systems.
- C) Nervous, Endocrine, and Reproductive systems. D) Excretory, Respiratory, and Nervous systems.

55. How many pairs of cranial nerves does a human possess?

- A) humans have 12 pairs. B) human have 10 pairs.
- C) humans have exactly 12 D) humans have 14 pairs.

56. The earthworm's body is covered in tiny, bristle-like structures called setae. What is their primary function and structural composition?

- A) They are sensory hairs made of keratin used for detecting vibrations.
- B) They are primarily for respiration, made of highly vascularized tissue.
- C) They are made of chitin and anchor the worm's body segments to the soil during peristaltic locomotion.
- D) They are calcified spines used defensively against predators.

57. If a healthy individual is bitten by an *Anopheles* mosquito carrying *Plasmodium*, what is the very first target organ the injected sporozoites migrate to via the blood stream?

- A) The Brain B) The Lungs C) The Red Blood Cells D) The Liver

58. Which of the following functions is NOT performed by the human liver?

- A) Synthesis of urea via the ornithine cycle.
- B) Secretion of digestive enzymes (like proteases) directly into the duodenum.
- C) Detoxification of drugs and alcohol.
- D) Storage of glucose as glycogen.

59. The basic rhythm of human respiration is involuntary and automatically controlled by respiratory centers located in the:

- A) Cerebral Cortex B) Cerebellum C) Medulla Oblongata and Pons D) Hypothalamus

60. The Sinoatrial (SA) node is considered the natural pacemaker of the heart because:

- A) It requires the largest amount of electrical input from the brain to fire.
- B) It generates electrical impulses faster than any other auto rhythmic tissue in the heart, driving the overall heart rate.
- C) It physically forces the blood into the ventricles.
- D) It connects the left and right sides of the heart.

61. In the human kidney, the vasa recta act as a countercurrent exchanger. What is its critical physiological role in urine concentration?

- A) It continuously pumps extra sodium into the urine to prevent dehydration.
- B) It supplies oxygen and nutrients to the renal medulla without washing out the high osmotic concentration gradient established by the Loop of Henle.
- C) It actively filters massive proteins directly from the blood.
- D) It synthesizes Antidiuretic Hormone (ADH) when blood pressure drops.

62. Which of the following physiological changes is a direct result of Parasympathetic nervous system activation?

- A) Dilation of the bronchioles in the lungs.
- B) Increased heart rate and contractility.
- C) Stimulation of digestive secretions and increased intestinal peristalsis.
- D) Dilation of the pupils to let in more light.

63. The olfactory receptors in the nasal cavity are unique among human sensory neurons because:

- A) They do not require a chemical stimulus to fire.
- B) They are capable of continuous regeneration from basal stem cells throughout a person's life.
- C) They bypass the brain entirely and connect directly to the spinal cord.
- D) They lack axons and release neurotransmitters directly into the air.

64. Which hormone is secreted by the alpha cells of the Islets of Langerhans, and what is its primary effect on target tissues?

- A) Insulin; it lowers blood glucose by promoting cellular uptake.
- B) Glucagon; it raises blood glucose by stimulating the liver to break down glycogen into glucose.
- C) Somatostatin; it inhibits the release of all digestive enzymes.
- D) Adrenaline; it triggers the fight-or-flight response.

65. During early pregnancy, the corpus luteum is maintained by Human Chorionic Gonadotropin (hCG). Which structure secretes this hormone?

- A) The anterior pituitary gland of the mother.
- B) The developing ovarian follicle.
- C) The trophoblast cells of the early developing embryo (blastocyst) and later the placenta.
- D) The endometrial lining of the uterus.

66. In the human digestive system, what is the primary function of the bicarbonate ions secreted in large volumes by the pancreas?

- A) To act as a buffer and neutralize the highly acidic chyme entering the duodenum from the stomach.
- B) To chemically denature complex proteins before enzyme action.
- C) To activate the absorption of Vitamin B12.
- D) To kill harmful bacteria ingested with food.

67. Which specific brain structure acts as the primary neuroendocrine link, directly controlling the secretions of the anterior and posterior pituitary gland?

- A) The Thalamus
- B) The Pineal Gland
- C) The Hypothalamus
- D) The Basal Ganglia

68. The visual defect known as Myopia (nearsightedness) occurs when light rays focus in front of the retina. This is typically corrected using which type of lens?

- A) A convex lens, to converge light rays.
- B) A concave lens, to diverge light rays before they enter the eye.
- C) A cylindrical lens, to correct an astigmatism.
- D) A bifocal lens.

69. In human the fertilization of sperm with ovum occurs at

- A) Vagina B) Penis C) Uterus D) Ampulla of oviduct

70. If a person has blood type O negative which antigens are present on the surface of their red blood cells?

- A) A antigens and Rh antigens B) B antigens only C) Both A and B antigens D) Neither A, B, nor Rh antigens

71. The human respiration occurs in

- A) Nose B) Lungs C) Cells D) Thorax

72. Which of the following is an example of an innate, non-specific physiological barrier in the human immune system?

- A) The production of specific IgG antibodies against the measles virus.
B) The low pH (high acidity) of gastric juice in the stomach destroying ingested pathogens.
C) The activation of memory T-cells.
D) The administration of a polio vaccine.

73. The Influenza virus frequently evades pre-existing herd immunity, requiring new vaccines annually. This is primarily due to 'antigenic shift'. What happens during an antigenic shift?

- A) Minor point mutations accumulate slowly over years.
B) The virus completely changes its RNA genome into DNA.
C) Two different strains of the virus infect the same host cell and reassort their genetic segments, creating a radically new viral subtype.
D) The virus learns to hide inside the host's nervous system.

74. Which biological enzyme, found abundantly in human tears, saliva, and sweat, acts as a chemical barrier by cleaving the peptidoglycan cell walls of Gram-positive bacteria?

- A) Lysozyme B) Pepsin C) Amylase D) Interferon

75. Hepatitis B and Hepatitis C are severe viral infections that primarily target the liver. Unlike Hepatitis A, which is transmitted via the fecal-oral route, how are Hepatitis B and C predominantly transmitted?

- A) Through inhalation of respiratory aerosols.
B) Through direct contact with infected blood or body fluids (e.g., shared needles, sexual contact).
C) By eating undercooked pork.
D) Through the bite of an infected tick.

76. Candidiasis, caused by the fungus *Candida albicans*, is typically described as an 'opportunistic infection'. What does this mean?

- A) The fungus only infects humans during the winter months.
B) The fungus is extremely aggressive and will infect any healthy individual upon contact.
C) The fungus is normally a harmless commensal organism in the body but causes disease only when the host's immune system is weakened or the normal bacterial flora is disrupted (e.g., by antibiotics).
D) The fungus can only infect humans after mutating inside a mosquito vector.

77. In modern sewage treatment, what is the primary biological process utilized during the 'Secondary Treatment' phase?

- A) The physical screening of large debris using metal grates.
B) The settling of heavy sludge and grit in large sedimentation tanks.
C) The aggressive use of aerobic microorganisms in aeration tanks to biologically degrade dissolved organic matter.
D) The heavy chemical chlorination of the water to kill all viruses.

78. Organ transplantation often faces the challenge of graft rejection. Which specific molecule on the surface of the transplanted organ's cells are recognized as foreign by the recipient's immune system, triggering the rejection?

- A) Ribosomes B) Major Histocompatibility Complex (MHC) proteins C) Cholesterol molecules D) Phospholipids

79. The Ramsar Convention is a major international environmental treaty. What specific type of ecosystem does this convention aim to protect and conserve globally?

- A) High-altitude alpine environments. B) Deep-sea coral reefs.

C) Wetlands critical for biodiversity, especially migratory water birds. D) Tropical rainforests.

80. Establishing National Parks, Wildlife Reserves, and Conservation Areas (such as Chitwan National Park in Nepal) are examples of which type of conservation strategy?

- A) Ex-situ conservation B) In-situ conservation C) Cryopreservation D) Genetic engineering

CHEMISTRY

81. Which has the maximum number of molecules?

- A) 7g of N_2 B) 2g of H_2 C) 16g of NO_2 D) 16g of O_2

82. Which set of quantum numbers (n, l, m, s) is not possible?

- A) 3, 2, -2, +1/2 B) 2, 0, 0, -1/2 C) 3, 3, 0, +1/2 D) 4, 1, +1, -1/2

83. Which group of elements exhibits the most pronounced paramagnetism?

- A) Alkali metals B) Noble gases C) Alkaline earth metals D) Transition metals

84. The dipole moment of CO_2 is zero, indicating that its structure is:

- A) Angular B) Linear C) Pyramidal D) Tetrahedral

85. Balancing redox equations in an acidic medium requires adding water to balance oxygen and what to balance hydrogen?

- A) OH^- ions B) H^+ ions C) Hydronium ions D) H_2 gas

86. The rate of diffusion of a gas is inversely proportional to the square root of its:

- A) Pressure B) Volume C) Temperature D) Molar mass

87. Deliquescent substances do what when exposed to the atmosphere?

- A) Lose water of crystallization B) Absorb moisture and dissolve in it C) Evaporate rapidly D) Change color

88. If Q_c (reaction quotient) > K_c , the reaction will:

- A) Proceed in the forward direction B) Proceed in the backward direction
C) Be at equilibrium D) Stop completely

89. A solution containing 1 mole of solute in 1 kg of solvent is a:

- A) 1 Molar solution B) 1 Molal solution C) 1 Normal solution D) 1 Formal solution

90. According to Bronsted-Lowry concept, a conjugate base is formed when an acid:

- A) Accepts a proton B) Donates a proton C) Accepts an electron pair D) Donates an electron pair

91. A catalyst increases the rate of reaction by:

- A) Increasing the activation energy B) Decreasing the enthalpy of reaction
C) Providing an alternative path with lower activation energy D) Increasing collision frequency

92. The standard hydrogen electrode (SHE) is assigned a potential of:

- A) 1.0 V B) 0.0 V C) -1.0 V D) Depends on temperature

93. In an adiabatic expansion of an ideal gas, the system:

- A) Absorbs heat B) Releases heat C) Neither absorbs nor releases heat D) Heats up

94. What particle is emitted in the decay $^{14}C \rightarrow ^{14}N$?

- A) Alpha particle B) Beta particle C) Gamma ray D) Positron

95. The bond order of the N_2 molecule is:

- A) 1 B) 2 C) 3 D) 4

96. The work done during the free expansion of an ideal gas in a vacuum is:

- A) Positive B) Negative C) Zero D) Infinite

97. An aqueous solution of a weak acid and a strong base has a pH at the equivalence point that is:

- A) Equal to 7 B) Less than 7 C) Greater than 7 D) Zero

98. Nascent hydrogen is more reactive than ordinary molecular hydrogen because it:

- A) Is hotter B) Has unpaired electrons (atomic state) C) Is a smaller molecule D) Contains impurities

99. Ammonia acts as a Lewis base because:

- A) It contains nitrogen B) It can release OH^- ions in water
C) Nitrogen has a lone pair of electrons to donate D) It forms hydrogen bonds

100. Which allotrope of carbon is thermodynamically the most stable?

- A) Diamond B) Graphite C) Fullerene D) Graphene

101. The main ore of Zinc is:

- A) Galena B) Bauxite C) Zinc blende (*Sphalerite*) D) Calamine

102. The term "zone refining" is based on the principle that:

- A) Impurities are more volatile than the pure metal B) Impurities are more soluble in the melt than in the solid metal
C) Metals have different boiling points D) Differential oxidation rates

103. Aqua regia is a mixture of concentrated acids in the ratio:

- A) 3 parts HCl : 1 part HNO_3 B) 3 parts HNO_3 : 1 part HCl
C) 1 part H_2SO_4 : 3 parts HCl D) 2 parts HNO_3 : 2 parts H_2SO_4

104. Rusting of iron is an electrochemical process involving the formation of:

- A) Fe_2O_3 B) Fe_3O_4 C) Hydrated iron(III) oxide D) Iron hydroxide

105. Which element is an essential macronutrient for plants but not for animals?

- A) Calcium B) Iron C) Boron D) Sodium

106. The formula of calomel is:

- A) HgCl_2 B) Hg_2Cl_2 C) HgSO_4 D) HgO

107. Acid rain is primarily caused by the emissions of:

- A) CO_2 and CO B) SO_2 and NO_2 C) CH_4 and CFCs D) O_3 and Dust

108. Functional isomers have the same molecular formula but different:

- A) Chain lengths B) Positions of the double bond C) Functional groups D) Spatial arrangement of atoms

109. The process of cracking involves:

- A) Polymerization of small molecules B) Breaking large hydrocarbon molecules into smaller ones
C) Adding oxygen to hydrocarbons D) Removing hydrogen to form double bonds

110. Markovnikov's rule applies to the addition of asymmetrical reagents to:

- A) Symmetrical alkenes B) Asymmetrical alkenes C) Alkanes D) Alkynes only

111. What is formed when chloroform is left open in the air and sunlight?

- A) Phosgene (highly poisonous) B) Carbon tetrachloride C) Chloropicrin D) Formic acid

112. The boiling point of ethanol is higher than that of dimethyl ether due to:

- A) Higher molecular mass B) Intermolecular hydrogen bonding
C) Dipole-dipole interactions D) Ionic bonding

113. Lucas test is used to distinguish between:

- A) Alkanes, alkenes, alkynes B) Primary, secondary, and tertiary alcohols

C) Aldehydes and ketones

D) Amines and amides

114. Tollen's reagent is chemically:

A) Alkaline potassium permanganate

B) Ammoniacal silver nitrate

C) Fehling solution A and B

D) Sodium bisulfite

115. Aldol condensation is undergone by aldehydes and ketones possessing at least one:

A) Alpha-hydrogen

B) Beta-hydrogen

C) Halogen atom

D) Double bond

116. Esterification of a carboxylic acid with an alcohol is carried out in the presence of:

A) Dilute base

B) Concentrated acid catalyst

C) Oxidizing agent

D) Reducing agent

117. Clemmensen reduction converts aldehydes and ketones into:

A) Alcohols

B) Alkanes

C) Carboxylic acids

D) Ethers

118. Aniline does not undergo Friedel-Crafts reaction because:

A) The ring is deactivated

B) Aniline forms a salt with the Lewis acid catalyst (AlCl_3)

C) Steric hindrance

D) It is an electrophile

119. The most common reagent to reduce nitrobenzene to aniline in the laboratory is:

A) LiAlH_4

B) Sn / HCl

C) $\text{Na} / \text{C}_2\text{H}_5\text{OH}$

D) H_2 / Pd

120. Grignard reagent reacts with formaldehyde (HCHO) to yield:

A) Primary alcohol

B) Secondary alcohol

C) Tertiary alcohol

D) Carboxylic acid

121. The functional group of an ether is:

A) $-\text{OH}$

B) $-\text{CHO}$

C) $-\text{O}-$

D) $-\text{COOH}$

122. Aspirin is chemically known as:

A) Methyl salicylate

B) Acetylsalicylic acid

C) Phenyl salicylate

D) Salicylaldehyde

123. Which effect stabilizes the carbocation intermediate in electrophilic substitution?

A) $-I$ effect

B) $+I$ effect

C) Resonance ($+R$ effect)

D) Both B and C

124. Dehydration of ethanol with concentrated H_2SO_4 at 170°C yields:

A) Diethyl ether

B) Ethane

C) Ethene

D) Ethanal

125. In continuous chemical plant processing, what is a primary advantage over batch processing?

A) Lower capital cost

B) Higher flexibility for multi-product lines

C) Steady-state operation yielding constant product quality

D) Easier to halt immediately

126. Which of the following is an example of an organochlorine pesticide?

A) Parathion

B) Malathion

C) DDT

D) Urea

127. The chemical name of baking soda is:

A) Sodium carbonate

B) Sodium bicarbonate

C) Sodium hydroxide

D) Calcium carbonate

128. In the Lassaigne test for Nitrogen, the Prussian blue color is due to the formation of:

A) Ferric ferrocyanide

B) Sodium cyanide

C) Potassium thiocyanate

D) Ferrous sulfate

129. Sublimation is a suitable separation technique when one component:

A) Is highly soluble in water

B) Vaporizes directly from solid to gas on heating

C) Has a high boiling point

D) Forms a precipitate

130. A primary standard substance used in volumetric analysis must:

A) Be highly hygroscopic

B) Have a low equivalent weight

C) Be extremely pure and stable in air

D) React very slowly

PHYSICS

131. If the energy E of a physical system depends on a parameter α according to the relation $E = (\alpha/\beta) e^{-(\alpha z / kBT)}$, where z is distance, kB is the Boltzmann constant, and T is absolute temperature, what are the dimensions of β ?

A) $[L^{-1}]$ B) $[L]$ C) $[M L T^{-1}]$

D) Dimensionless

132. A block of mass m is placed on a rough inclined plane of inclination θ . The coefficient of static friction between the block and the plane is μ_s (where $\mu_s < \tan \theta$). A horizontal force F is applied to the block, pushing it against the incline to prevent it from sliding down. What is the minimum magnitude of F required?

A) $mg (\sin \theta - \mu_s \cos \theta) / (\cos \theta + \mu_s \sin \theta)$ B) $mg (\sin \theta + \mu_s \cos \theta) / (\cos \theta - \mu_s \sin \theta)$ C) $mg (\sin \theta - \mu_s \cos \theta)$ D) $mg \tan \theta$

133. Two capillary tubes of lengths L and $2L$, and radii R and $2R$ respectively, are connected in series. An ideal viscous liquid flows through them steadily. What is the ratio of the pressure drop across the first tube to the pressure drop across the second tube?

A) 8 : 1

B) 1 : 8

C) 16 : 1

D) 1 : 1

134. A solid uniform sphere of mass M and radius R rolls without slipping down an inclined plane, starting from rest at a vertical height h . What is its purely translational kinetic energy when it reaches the bottom of the incline?

A) $(5/7) Mgh$ B) $(2/7) Mgh$ C) $(1/2) Mgh$ D) Mgh

135. The gravitational potential energy of a body of mass m at the surface of the Earth is $-mgR_e$ (where R_e is the radius of the Earth). What is the minimum work required to move this body from the Earth's surface to a height $h = R_e / 2$?

A) $(1/3) mgR_e$ B) $(1/2) mgR_e$ C) $(2/3) mgR_e$ D) mgR_e

136. A wire of original length L and cross-sectional area A is stretched by an applied force F . If the Young's modulus of the wire material is Y , what is the elastic potential energy density (energy per unit volume) stored in the stretched wire?

A) $F^2 / (2A^2Y)$ B) $F^2 / (2AY)$ C) $F / (2AY)$ D) $(1/2) (YA / F^2)$

137. A block of mass m is placed on a frictionless horizontal surface and is attached to two fixed walls by two unstretched springs of spring constants k_1 and k_2 on opposite sides. If the mass is displaced slightly along the line of the springs, what is the effective spring constant for the resulting simple harmonic motion?

A) $k_1 + k_2$ B) $(k_1 k_2) / (k_1 + k_2)$ C) $|k_1 - k_2|$ D) $\sqrt{k_1 k_2}$

138. A projectile is fired from the ground with an initial velocity u at an angle θ to the horizontal. Assuming no air resistance, what is the radius of curvature of its parabolic trajectory at the absolute highest point of its flight?

A) $(u^2 \cos 2\theta) / g$ B) $(u^2 \sin 2\theta) / g$ C) $u^2 / (g \cos \theta)$ D) u^2 / g

139. A large open tank has a small hole at a depth h below the free surface of a liquid contained within it. The liquid streams out of the hole. If the entire tank (including the liquid) is allowed to fall freely under gravity, what will be the velocity of efflux of the liquid relative to the tank?

A) 0

B) $\sqrt{2gh}$ C) $\sqrt{gh}$ D) $2gh$

140. A particle of mass m moving with velocity v undergoes a perfectly elastic, head-on collision with a stationary particle of mass M . What fraction of the initial kinetic energy of the incident particle is transferred to the stationary particle during the collision?

A) $4mM / (m+M)^2$ B) $2mM / (m+M)^2$ C) $m / (m+M)$ D) $M / (m+M)$

141. The absolute zero of temperature is the temperature at which:

- A) Water freezes B) Molecular motion ceases C) Volume of gas becomes infinite D) Pressure becomes maximum

142. Two rods of the same dimensions but thermal conductivities K_1 and K_2 are connected in series. The equivalent thermal conductivity is:

- A) $(K_1 + K_2)/2$ B) $2K_1K_2/(K_1 + K_2)$ C) $K_1 + K_2$ D) $\sqrt{K_1K_2}$

143. The first law of thermodynamics is a statement of:

- A) Conservation of mass B) Conservation of momentum C) Conservation of energy D) Conservation of heat

144. In an adiabatic process, the quantity that remains constant is:

- A) Temperature B) Pressure C) Volume D) Entropy (Heat)

145. Latent heat of fusion of ice is:

- A) 540 cal/g B) 80 cal/g C) 1 cal/g°C D) 4.2 J/g

146. A perfect black body is one which:

- A) Absorbs all incident radiation B) Reflects all incident radiation
C) Transmits all radiation D) Is perfectly black in color

147. Mayer's formula for an ideal gas is:

- A) $C_p/C_v = \gamma$ B) $C_p + C_v = R$ C) $C_p - C_v = R$ D) $C_v - C_p = R$

148. The pitch of a sound wave depends upon its:

- A) Amplitude B) Frequency C) Speed D) Intensity

149. Beats are the result of:

- A) Diffraction B) Polarization C) Interference D) Reflection

150. When light travels from a rarer to a denser medium, its:

- A) Velocity increases B) Wavelength increases
C) Frequency remains constant D) Frequency increases

151. A convex lens of focal length 20 cm is cut into two equal halves along its principal axis. The focal length of each half is:

- A) 10 cm B) 20 cm C) 40 cm D) Infinity

152. The splitting of white light into its component colors is called:

- A) Dispersion B) Diffraction C) Interference D) Scattering

153. Optical fibers work on the principle of:

- A) Refraction B) Scattering C) Total Internal Reflection D) Dispersion

154. Brewster's law states that at the polarizing angle i_p :

- A) $\mu = \sin i_p$ B) $\mu = \tan i_p$ C) $\mu = \cos i_p$ D) $\mu = \cot i_p$

155. In a Young's double slit experiment, if the source is moved closer to the slits, the fringe width:

- A) Increases B) Decreases C) Remains unchanged D) Becomes zero

156. The specific resistance of a wire depends upon:

- A) Its length B) Its cross-sectional area C) Its material D) Its mass

157. A galvanometer can be converted into an ammeter by connecting a:

- A) High resistance in series B) Low resistance in parallel (shunt)
C) High resistance in parallel D) Low resistance in series

158. According to Kirchhoff's Voltage Law (KVL), the algebraic sum of potential drops in a closed loop is:

- A) Infinite B) Zero C) Equal to current D) Equal to resistance

159. The angle of dip at the magnetic equator is:

- A) 0° B) 45° C) 90° D) 180°

160. The SI unit of magnetic flux is:

- A) Tesla B) Weber C) Henry D) Ampere/meter

161. Motional emf induced in a conductor of length l moving with velocity v perpendicular to a magnetic field B :

- A) Bvl B) Bv/l C) Bl/v D) vl/B

162. An inductor stores energy in the form of a:

- A) Electric field B) Magnetic field C) Both A and B D) Neither

163. The power factor of a purely resistive circuit is:

- A) 0 B) 0.5 C) 1 D) Infinity

164. Alternating current is converted to direct current by:

- A) Transformer B) Motor C) Rectifier D) Generator

165. The Coulomb force between two charges is an example of a:

- A) Strong nuclear force B) Weak force C) Central force D) Non-conservative force

166. The work done in moving a charge q along an equipotential surface is:

- A) qV B) q/V C) Zero D) Infinite

167. Two capacitors of $2\mu\text{F}$ and $4\mu\text{F}$ are connected in parallel. Their equivalent capacitance is:

- A) $1.33\mu\text{F}$ B) $6\mu\text{F}$ C) $8\mu\text{F}$ D) $2\mu\text{F}$

168. The relative permittivity (dielectric constant) of metals is:

- A) Zero B) One C) Greater than one but finite D) Infinite

169. In Millikan's oil drop experiment, an oil drop of mass m and charge q is observed to fall with a terminal velocity v_1 under the influence of gravity alone. When a vertical upward electric field E is applied, the drop rises with a new terminal velocity v_2 . Assuming the buoyant force of air is negligible, the charge q on the drop is given by:

- A) mg / E B) $mg(v_1 + v_2) / (E v_1)$ C) $mg(v_1 - v_2) / (E v_1)$ D) $mg v_2 / (E v_1)$

170. According to Einstein's photoelectric equation, the K.E of emitted photoelectrons depends linearly on:

- A) Intensity of light B) Wavelength of light C) Frequency of light D) Time of illumination

171. The Balmer series of the hydrogen spectrum lies in the:

- A) Ultraviolet region B) Visible region C) Infrared region D) Microwave region

172. According to Hubble's law, a distant galaxy is observed to be receding from Earth at a velocity of 3×10^4 km/s. Assuming the value of the Hubble constant H_0 is 75 km/s/Mpc, what is the estimated distance of this galaxy from Earth?

- A) 200 Mpc B) 300 Mpc C) 400 Mpc D) 500 Mpc

173. The relation between decay constant λ and mean life τ is:

- A) $\tau = \lambda$ B) $\tau = 1/\lambda$ C) $\tau = 0.693/\lambda$ D) $\tau = \lambda^2$

174. In a nuclear reactor, control rods are made of:

- A) Uranium B) Cadmium C) Graphite D) Heavy water

175. The energy equivalent of 1 kg of mass is approximately:

- A) 9×10^{16} J B) 3×10^8 J C) 9×10^8 J D) 10^{10} J

176. A Zener diode is primarily used as a:

- A) Rectifier B) Amplifier C) Voltage regulator D) Oscillator

177. The Boolean expression $Y = A + B$ represents a:

- A) NAND gate B) NOR gate C) AND gate D) OR gate

178. Pure silicon at absolute zero temperature behaves as:

- A) A good conductor B) A semiconductor C) An insulator D) A superconductor

179. Transistors are basically:

- A) Voltage operated devices B) Current operated devices C) Power operated devices D) Resistance operated devices

180. Heisenberg's uncertainty principle states that it is impossible to simultaneously measure accurately:

- A) Position and momentum B) Energy and mass C) Time and space D) Charge and mass

MAT

181. OBFUSCATE : CLARITY?

- A) Ameliorate : Improvement B) Mollify : Anger C) Exacerbate : Severity D) Capitulate : Surrender

182. All glops are tranks. Some tranks are zinds. No zind is a frelk. Which logically follows beyond reasonable doubt?

- I. Some glops are zinds. II. No glop is a frelk. III. Some tranks are not frelks.
A) Only I follows B) Only II follows C) Only III follows D) None follows

183. Identify the word that does NOT belong with the others.

- A) Ephemeral B) Evanescent C) Transient D) Pertinacious

184. If all blips are blops, and no blops are gloops, which must be true?

- A) All gloops are blips. B) No blips are gloops. C) Some blips are gloops. D) Some gloops are blops.

185. City council's toll proposal on highways is flawed; studies show traffic diverts to local roads, increasing congestion. Which weakens this argument?

- A) Toll revenue will improve transit eventually. B) Local roads have been widened.
C) Tolls are active only in rush hours. D) Drivers prefer paying tolls over slower local routes.

186. What number logically follows: 2, 5, 14, 41, 122, ?

- A) 365 B) 367 C) 361 D) 363

187. Pipe A fills a tank in 12h, B empties in 18h. Both open simultaneously. After 6h, B is permanently closed. Total time from start to fill tank?

- A) 10 hours B) 12 hours C) 14 hours D) 16 hours

188. A bag contains 4 red, 5 green, 6 blue balls. Three drawn at random. Probability exactly two are green?

- A) 24/91 B) 20/91 C) 16/91 D) 12/91

189. A man's age is a two-digit number. Reversed, it gives his age 18 years ago. Sum of digits is 10. How old is he?

- A) 46 B) 64 C) 73 D) 82

190. In an election, 10% didn't vote, 10% of cast votes were invalid. Winner got 54% of valid votes, won by 1,620 majority. Total enrolled voters?

- A) 25,000 B) 30,000 C) 35,000 D) 40,000

191. Next in sequence: AZ, CX, FU, JQ, ?

- A) OL B) OK C) PK D) PL

192. Missing number: 3, 4, 10, 33, 136, ?

- A) 685 B) 680 C) 695 D) 690

193. A, B, C, D, E sit facing north. C is between A and E. D is immediate right of E. B is extreme left. Who is exactly in the middle?

- A) A B) C C) E D) D

194. Code "COMPUTER" -> "RFUVQNPC". Code "MEDICINE" -> ?

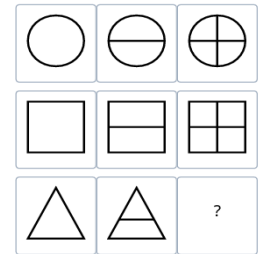
- A) EOJDJEFM B) EOJDEJFM C) MFEJDJOE D) MFEDJJOE

195. If 3rd of a month is Wednesday, what day is 4th day after the 21st?

- A) Monday B) Tuesday C) Wednesday D) Thursday

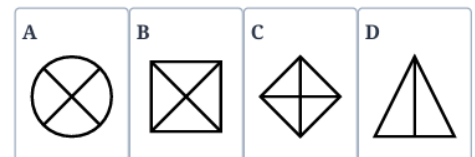
196. Analyze the matrix progression. What figure completes the pattern?

- A) A triangle with only a vertical line
B) triangle with a horizontal and vertical crossline
C) A circle with a horizontal and vertical crossline
D) A square with a diagonal line



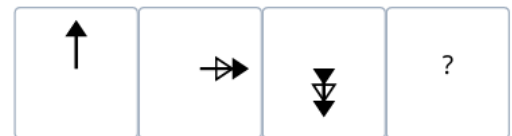
197. 17. Identify the odd visual figure out of the four given options.

- A) Figure A
B) Figure B
C) Figure C
D) Figure D



198. A sequence of rotation and element addition occurs below. What is the next logical figure?

- A) Arrow pointing Left, with 4 arrowheads (alternating fill).
B) Arrow pointing Top-Left, with 3 arrowheads
C) Arrow pointing Left, with 3 arrowheads (all black)
D) Arrow pointing Top-Right, with 4 arrowheads (alternating fill)



199. A transparent square sheet has a line drawn on it. If the sheet is folded perfectly along the dotted vertical center line (from right to left), what will the resulting image look like?

- A) A single diagonal line from top-left to bottom-right.
B) An 'X' shape crossing the entire folded rectangle.
C) A 'V' shape occupying the folded rectangular area
D) Two parallel diagonal lines



200. Which of the following shows the correct MIRROR IMAGE of the shape provided below (assuming the mirror is placed vertically on the right)?

- A) A shape identical to the original, with the dot shifted to the right.
B) A shape pointing leftwards: points="80,20 20,20 40,50 20,80 80,80 60,50" with dot at cx=65.
C) An inverted shape pointing upwards with dot at bottom.
D) A shape identical to the original but rotated 90 degrees.

